



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO AGNI FUEL

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

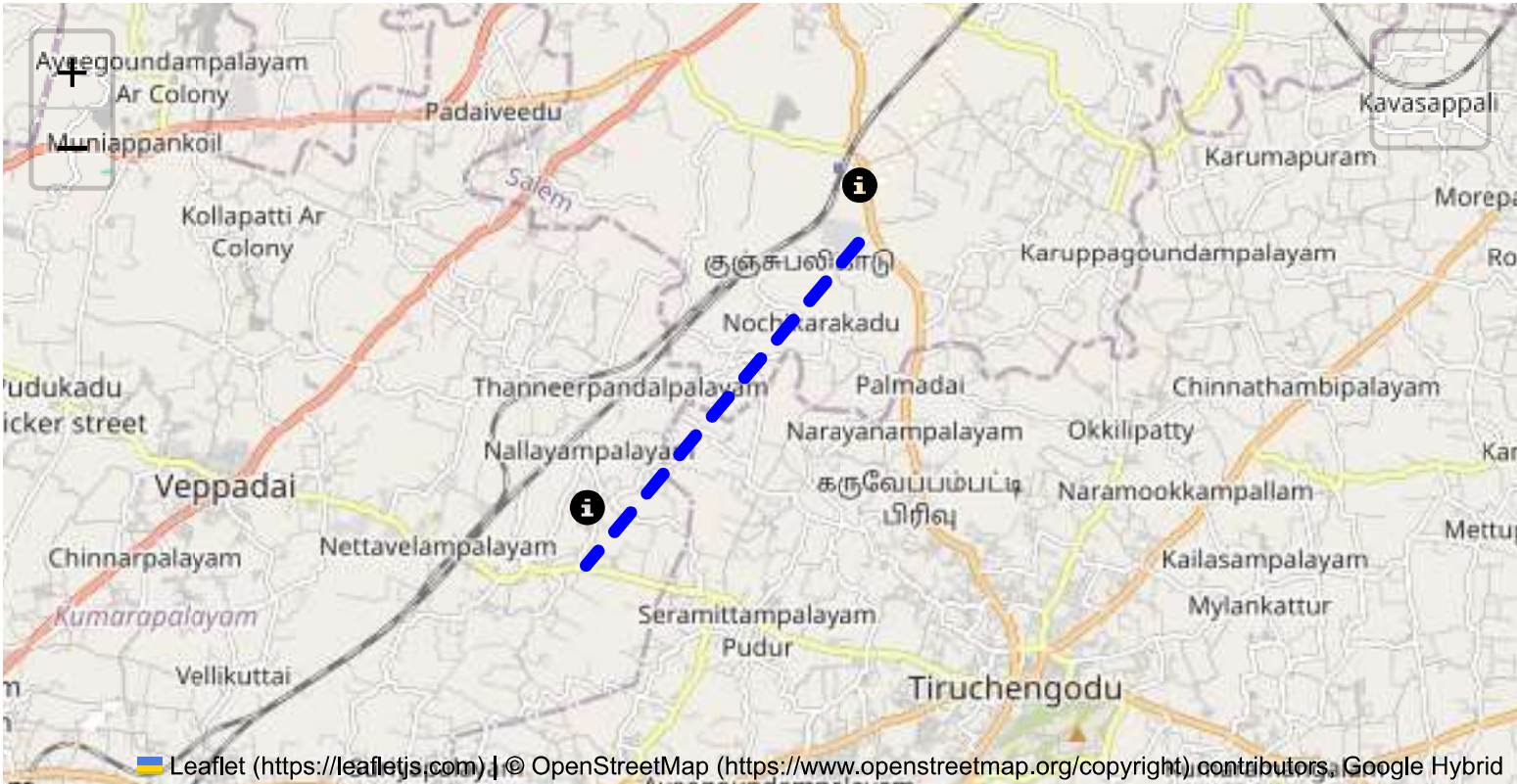
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

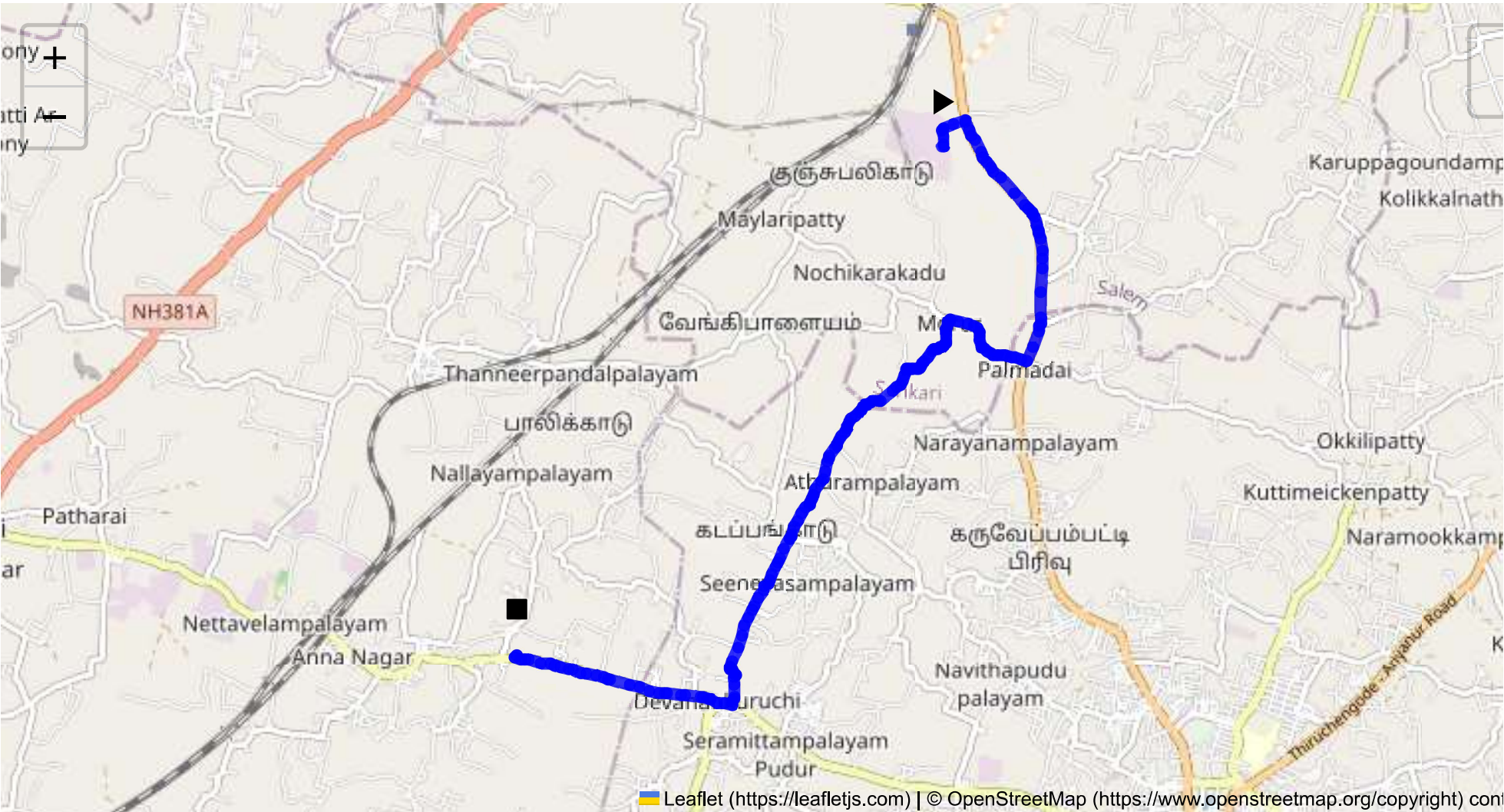
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 10.47 km
Estimated Duration: 0.3 hours
Adjusted Duration (Heavy Vehicle): 0.4 hours
Start: (11.4381, 77.8734)
End: (11.394823, 77.836332)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Route Overview

The route from Sangagiri to Andikadu travels approximately 10.47 kilometers, primarily along smaller regional roads. The main connecting road is likely rural with potential intersections connecting local

villages and towns.

2. Weather Conditions and Hazards

- **Typical Weather:** The region experiences a tropical climate with distinct wet and dry seasons. The monsoon season (June to September) often leads to heavy rainfall.
- **Weather-Related Hazards:** During monsoons, expect potential flooding or waterlogging, which could affect road conditions. Dry seasons may have occasional heat waves that could impact vehicle performance.

3. Traffic Patterns

- **Peak Hours:** Morning (8–10 AM) and evening (5–7 PM) are typical peak periods.
- **Congestion-Prone Areas:** The vicinity of Sangagiri and the approach to Komarapalayam might experience moderate congestion due to local traffic, especially at intersections or markets.

4. Road Quality and Infrastructure

- **Road Quality:** Varies from well-maintained to potentially bumpy in rural stretches. Expect occasional potholes, particularly post-monsoon.
- **Infrastructure:** Road signs and markings may be limited in rural areas. Be cautious near villages where pedestrian and livestock crossings are common.

5. Emergency Alternative Routes

- **Suggested Routes:** In case of blockage, consider routes toward Erode or Salem, which have better infrastructure and emergency facilities.

6. Local Regulations on Hazardous Materials

- **Transport Regulations:** Tamil Nadu has stringent regulations on hazardous material transport, often requiring specific permits. Restrictions may apply through town centers during peak hours.

7. Historical Incidents

- **Incident Overview:** There are sporadic reports of road accidents involving heavy vehicles in the region, often due to road conditions and human error, but no major incidents involving hazardous materials are noted along this specific route.

8. Environmental Considerations

- **Sensitive Areas:** Be aware of agricultural fields alongside roads. Avoid contaminating nearby water bodies or farmlands. Local wildlife may be active crossing the road.

9. Communication Coverage

- **Coverage Analysis:** Generally good along the route. However, some rural stretches may experience intermittent connectivity or dead zones, particularly in lower-lying areas.

10. Emergency Response Times

- **Estimated Times:** Given the proximity to towns like Komarapalayam and Tiruchengode, emergency services could take 15-30 minutes, depending on road conditions and time of day.

11. Overall Summary of Risk Assessment

The Sangagiri to Andikadu route presents moderate risks typical of rural Indian roads. The primary concerns lie in weather-related disruptions during the monsoon, potential congestion near population centers, and intermittent communication coverage. Proper adherence to local transport regulations for hazardous materials, paired with pre-planning and alternate routes, will mitigate most operational risks. Ensure that drivers are trained in navigating rural roads and prepared for delayed responses in emergencies.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	Medium	11.43971, 77.87348	30 KM/Hr
3	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
4	Blind Spot	Blind Spot	11.41981, 77.88078	10 KM/Hr
5	Turn	Medium	11.42102, 77.87680	30 KM/Hr
6	Turn	Medium	11.42231, 77.87666	30 KM/Hr
7	Blind Spot	Blind Spot	11.42333, 77.87394	10 KM/Hr
8	Turn	Medium	11.42129, 77.87359	30 KM/Hr
9	Turn	Medium	11.42090, 77.87311	30 KM/Hr
10	Turn	High	11.41917, 77.87022	15 KM/Hr
11	Turn	Medium	11.41642, 77.86800	30 KM/Hr
12	Turn	High	11.40478, 77.86020	15 KM/Hr
13	Turn	High	11.40471, 77.86004	15 KM/Hr
14	Turn	Medium	11.39360, 77.85502	30 KM/Hr
15	Turn	Medium	11.39325, 77.85538	30 KM/Hr
16	Turn	High	11.39047, 77.85509	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
17	Blind Spot	Blind Spot	11.39447, 77.83631	10 KM/Hr
18	Turn	High	11.39466, 77.83635	15 KM/Hr

Route Photos of Risky Spots



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.41981, 77.88078



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.42333, 77.87394



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.41917, 77.87022



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.41642, 77.86800



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.40478, 77.86020



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.40471, 77.86004



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39360, 77.85502



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.39325, 77.85538



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.39047, 77.85509



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.39447, 77.83631



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.39466, 77.83635

